

not clear how to interpret the size of any employment effect found for those groups.⁶ Third, the localized focus around the minimum wage often improves the precision of estimates by filtering out random shocks to jobs in the upper part of the wage distribution.⁷

We use the estimated $\alpha_{\tau k}$ from equation (1) to calculate the change in employment throughout the wage distribution in response to the policy. The change in the number of jobs (per capita) paying below the new minimum wage between event date -1 and τ can be calculated as $\sum_{k=-4}^{-1} \alpha_{\tau k} - \sum_{k=-4}^{-1} \alpha_{-1k}$. To be clear, this is a difference-in-differences estimate, as it nets out the change in the counterfactual distribution implicitly defined by the regression equation (1). Analogously, the change in the number of jobs (per capita) paying between the minimum wage and $\bar{W}$ is $\sum_{k=0}^{\bar{W}-MW} \alpha_{\tau k} - \sum_{k=0}^{\bar{W}-MW} \alpha_{-1k}$. For our baseline estimates, we set $\bar{W} = MW + 4$.⁸ We define the excess jobs at or above the minimum wage as $\Delta a_{\tau} = \frac{\sum_{k=0}^4 \alpha_{\tau k} - \sum_{k=0}^4 \alpha_{-1k}}{EPOP_{-1}}$, and the missing jobs below as $\Delta b_{\tau} = \frac{\sum_{k=-4}^{-1} \alpha_{\tau k} - \sum_{k=-4}^{-1} \alpha_{-1k}}{EPOP_{-1}}$. By dividing the employment changes by $\overline{EPOP}_{-1}$, the sample average employment-to-population ratio in treated states during the year (four quarters) prior to treatment, we normalize the excess and missing jobs by the pretreatment total employment. The Δa_{τ} and Δb_{τ} values plot out the evolution of excess and missing jobs over event time τ . We also report the excess and missing employment estimates averaged over the five years following the minimum wage increase, $\Delta b = \frac{1}{5} \sum_{\tau=0}^4 \Delta b_{\tau}$ and $\Delta a = \frac{1}{5} \sum_{\tau=0}^4 \Delta a_{\tau}$.

Given our normalization, $\Delta e = \Delta a + \Delta b$ represents the estimate for the percentage change in total employment due to the

6. In Online Appendix Table A.1 we demonstrate that the standard approach, which looks at the wage and employment effects aggregated over the entire wage distribution, fails to produce positive and statistically significant wage effects in most cases. This indicates that the standard approach fails to capture the program effect of the minimum wage for these subgroups. At the same time, our estimates focused on low-wage jobs always produce sizable and significant wage effects. The own-wage elasticity of employment estimated using minimum wage variation is effectively a Wald-IV estimate; hence the lack of a strong “first stage” means estimates are biased toward the OLS estimate obtained by naively regressing employment on wages (Bound, Jaeger, and Baker 1995).

7. Online Appendix Table A.2 confirms that the standard errors tend to be lower when we consider counts of low-wage jobs compared with an approach using total number of jobs.

8. Online Appendix Table A.5 shows that the results are robust to higher cutoffs.

minimum wage increase. We refer to this estimate as “event-based bunching” or EB-bunching estimates to highlight that we are (i) using an event-based difference-in-differences design and (ii) estimating the excess and missing jobs locally around the bunching in the distribution at the minimum wage.

If we divide Δe by the percentage change in the minimum wage averaged across our events, $\% \Delta MW$, we obtain the employment elasticity with respect to the minimum wage:

$$\frac{\% \Delta \text{Total Employment}}{\% \Delta MW} = \frac{\Delta a + \Delta b}{\% \Delta MW}.$$

We define the percentage change in affected employment as the change in employment divided by the (sample average) share of the workforce earning below the new minimum wage the year before treatment, $\bar{b}_{-1}$:⁹

$$\% \Delta \text{Affected Employment} = \% \Delta e = \frac{\Delta a + \Delta b}{\bar{b}_{-1}}.$$

We also use the estimated coefficients to compute the percentage change in the average hourly wage for affected workers. We calculate the average wage by taking the ratio of the total wage bill collected by workers below the new minimum wage to the number of such workers. Prior to treatment, it is equal to $\bar{w}_{-1} = \frac{\bar{wb}_{-1}}{\bar{b}_{-1}}$. Here the wage bill, $\bar{wb}_{-1}$, and the number of workers earning below the new minimum wage just prior to the increase, $\bar{b}_{-1}$, are averages for the full sample of events. The minimum wage increase causes both the wage bill and employment to change. The new average wage in the post-treatment period is equal to $w = \frac{\bar{wb}_{-1} + \Delta wb}{\bar{b}_{-1} + \Delta e}$.¹⁰ Therefore,

9. Notice that we divide by the actual share of the workforce and not by the change in it. As we pointed out earlier, these two are not the same if there is imperfect compliance, imperfect coverage, or measurement error in wages. Although both divisions are meaningful, dividing by the actual share is the more policy-relevant elasticity. This is because policy makers can calculate the actual share of workers at the new minimum wage and use the estimates presented in this article. However, the change in the jobs below the new minimum wage is only known after the minimum wage increase, so it cannot be used for a prospective analysis of the policy's impact.

10. The change in wage bill can be written as a function of our regression coefficients as follows. Averaging the coefficients over the five-year post-treatment window, $\alpha_k = \frac{1}{5} \sum_{\tau=0}^4 \alpha_{\tau,k}$, we can write $\Delta wb = \sum_{k=-3}^4 (k + \bar{MW}) \cdot (\alpha_k - \alpha_{-1k})$, where

the percentage change in the average wage of affected workers is given by:

$$(2) \quad \% \Delta w = \frac{w}{\bar{w}_{-1}} - 1 = \frac{\frac{\overline{wb}_{-1} + \Delta wb}{\bar{b}_{-1} + \Delta e}}{\frac{\overline{wb}_{-1}}{\bar{b}_{-1}}} - 1 = \frac{\% \Delta wb - \% \Delta e}{1 + \% \Delta e}.$$

The percentage change in the average wage is obtained by taking the difference in percentage change in wage bill and employment, and dividing by the retained employment share. This formula implicitly assumes the average wage change of those workers exiting or entering due to the policy is the same as the average wage change of affected workers who remain employed.

Finally, armed with the changes in employment and wages for affected workers, we estimate the employment elasticity with respect to own-wage (or the “labor demand elasticity” in a competitive market):

$$\frac{\% \Delta \text{Affected Employment}}{\% \Delta \text{Affected Wage}} = \frac{1}{\% \Delta w} \frac{\Delta a + \Delta b}{\bar{b}_{-1}}.$$

We calculate the standard errors for this elasticity using the delta method.

Although our use of wage-bin-by-state-by-quarter data is useful for decomposing the employment changes by bins relative to the minimum wage, our employment and wage elasticities do not rely on this binning. To clarify this point, we show results from a simpler method that estimates a regression using state-by-quarter data, where the outcomes are the (per capita) number of jobs or total wage bill under, say, \$15/hour, and the event indicators are just by state and quarter. We show below that the resulting employment and wage estimates (and standard errors) are very similar when using this simpler method.

II.C. Data and Sample Construction

We use the individual-level NBER Merged Outgoing Rotation Group of the Current Population Survey for 1979–2016 (CPS) to

$\overline{MW}$ is (approximately) the sample average of the new minimum wage. We say approximately because k is based on \$1 increments, and so $\overline{MW}$ is calculated as the sample mean of $[MW, MW + 1)$.

calculate quarterly, state-level distributions of hourly wages. For hourly workers, we use the reported hourly wage, and for other workers we define the hourly wage to be their usual weekly earnings divided by usual weekly hours. We do not use any observations with imputed wage data to minimize the role of measurement error.¹¹ There are no reliable imputation data for January 1994 through August 1995, so we exclude this entire period from our sample. Our available sample of employment counts therefore spans 1979q1 through 1993q4 and 1995q4 through 2016q4.¹²

We deflate wages to 2016 dollars using the CPI-U-RS and for a given real hourly wage assign its earner a \$0.25 wage bin w running from \$0.00 to \$30.00.¹³ For each of these 117 wage bins we collapse the data into quarterly, state-level employment counts E_{swt} using the person-level ORG sampling weights. We use estimates for state-level population aged 16 and over, N_{st} , from the CPS-MORG (which in turn is based on the census), as the denominator for constructing per capita counts. Our primary sample includes all wage earners and the entire state population, but we also explore the heterogeneity of our results using different subgroups, where the bite of the policy varies.

The aggregate state-quarter-level employment counts from the CPS are subject to sampling error, which reduces the precision of our estimates. To address this issue, we benchmark the CPS aggregate employment-to-population ratio to the implied employment-to-population ratio from the Quarterly Census

11. The NBER CPS merged ORG data are available at <http://www.nber.org/morg/>. Wage imputation status markers in the CPS vary and are not comparable across time. In general we follow [Hirsch and Schumacher \(2004\)](#) to define wage imputations. During 1979–1988 and September 1995–2016, we define wage imputations as records with positive BLS allocation values for hourly wages (for hourly workers) and weekly earnings or hours (for other workers). For 1989–1993, we define imputations as observations with missing or zero “unedited” earnings but positive “edited” earnings (which we also do for hours worked and hourly wages).

12. In general, there has been an increase in the rate of imputation over time. However, in [Online Appendix Table A.3](#) and [Online Appendix Figure A.2](#), we show that minimum wage raises are not systematically related to changes in the imputation rate. Event study estimates for the effect of minimum wages on the imputation rate show no substantial or statistically significant change three years before and five years after the treatment.

13. We assign all wages between \$0 and \$1 to a single bin and all wages above \$30 to the \$30 bin. The resulting 117 wage bins are (0.00, 1.25), [1.25, 1.50), ..., [29.75, 30.00), [30, ∞).